

NEUROSCOPE

Learning Generalizable Representations of Human Decision-Making Using Self-Supervised Learning

Executive Summary

Human decision-making is traditionally modeled using computational frameworks such as Reinforcement Learning (RL), Drift Diffusion Models (DDM), Bayesian inference, and, more recently, information-theoretic approaches. These models estimate latent parameters—such as learning rate, exploration tendency, or decision thresholds—to explain observed behavior.

Recent advances have improved these models by allowing parameters to vary over time or by using neural networks to estimate them more accurately. However, these approaches still assume that **human behavior follows a predefined computational model**.

This project proposes a different approach.

Instead of fitting human behavior to existing theories, **NEUROSCOPE** aims to learn a **general latent representation of decision-making directly from behavioral data** using self-supervised learning. The goal is not to replace existing cognitive models, but to determine whether learned behavioral representations generalize across multiple cognitive tasks better than task-specific computational parameters.

Research Gap

The recent literature shows three major trends:

- **Dynamic RL models** estimate time-varying learning parameters rather than fixed ones.
- **Information-theoretic models** infer learning strategies using entropy and mutual information rather than traditional regression.
- **Meta-learning frameworks** suggest humans adapt their learning strategies over time rather than relying on fixed strategies.

Despite these advances, current approaches share three limitations:

1. They rely on **predefined computational models** (RL, Bayesian, DDM).
2. They are generally **task-specific**, with limited evaluation across multiple cognitive domains.
3. They summarize behavior using a **small number of latent parameters**, potentially overlooking richer behavioral patterns.

No current framework has demonstrated that a **single learned behavioral representation** can capture human decision-making across diverse cognitive tasks without assuming a specific computational model.

Proposed Idea

NEUROSCOPE will collect rich behavioral sequences from participants performing multiple decision-making tasks, including probabilistic learning, risk preference, delay discounting, rule discovery, and social decision-making.

Rather than estimating predefined parameters, a **self-supervised Transformer encoder** will learn compact behavioral embeddings directly from these sequences.

The learned representation will then be evaluated on three questions:

1. **Cross-task generalization:** Can a representation learned from some tasks predict behavior in unseen tasks?
2. **Behavioral consistency:** Are these representations stable across repeated testing?
3. **Comparison with classical models:** Do learned representations explain behavior better than RL or DDM parameters alone?

Expected Contribution

This project aims to contribute a **general representation-learning framework for behavioral science**, rather than another task-specific decision model.

Expected outcomes include:

- A latent embedding space describing human decision behavior.
- Quantitative evaluation of cross-task behavioral generalization.
- Direct comparison between learned representations and classical computational models.
- An open behavioral dataset and reproducible AI pipeline.

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